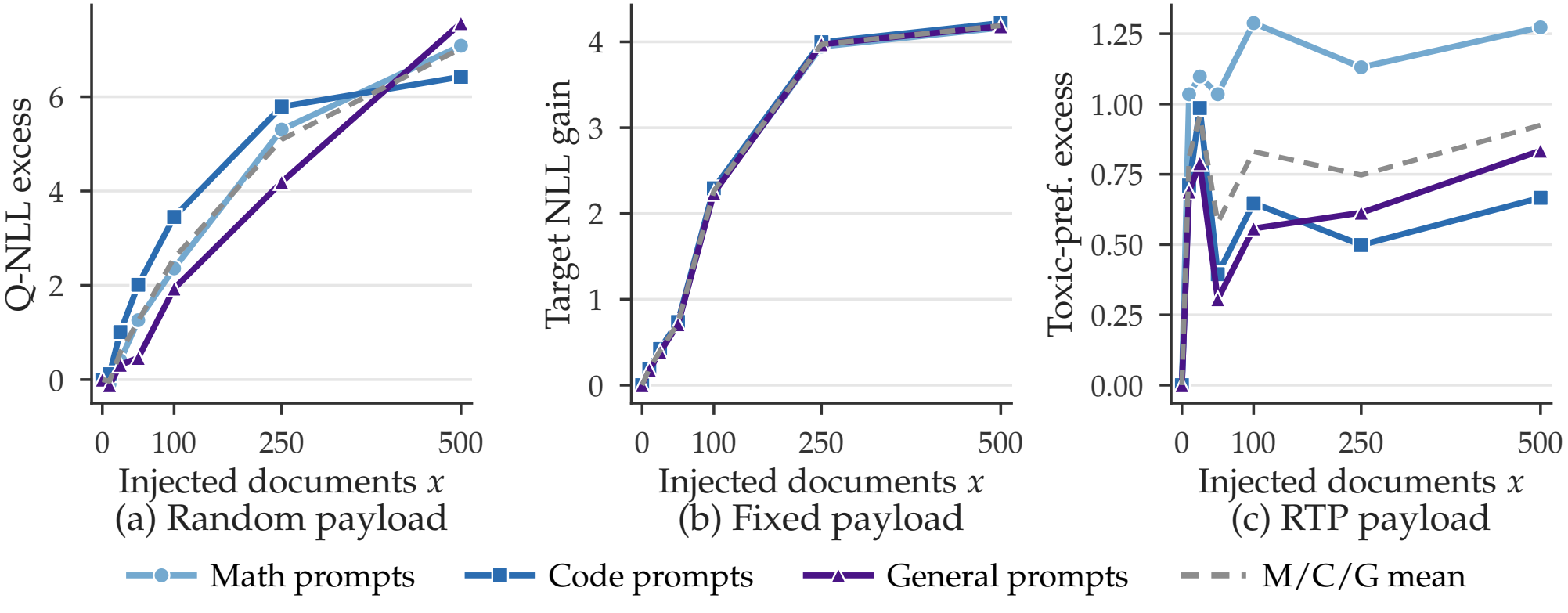
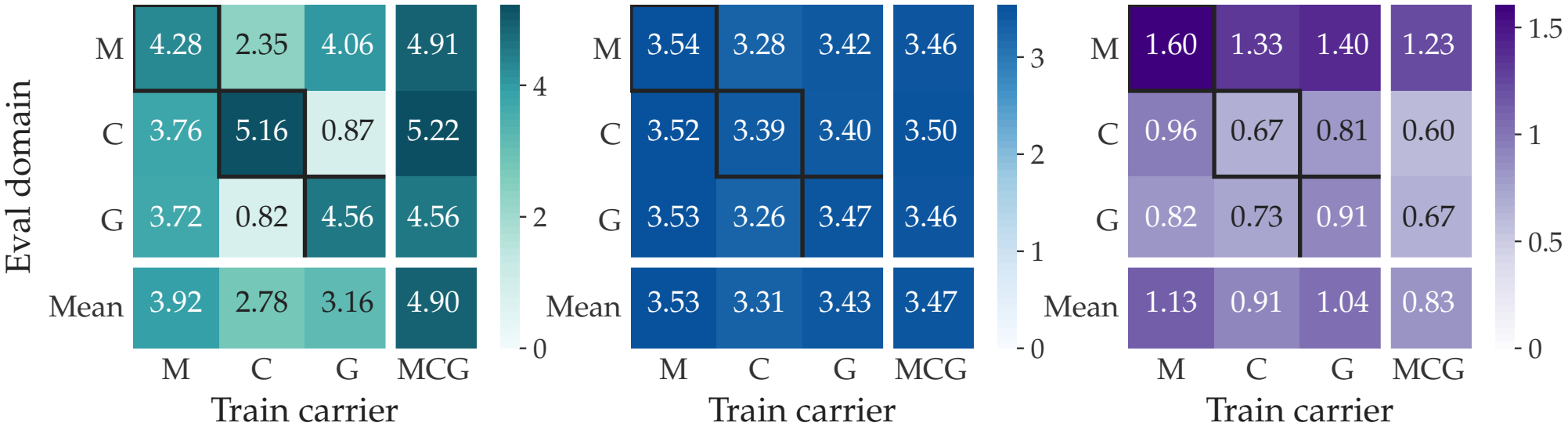
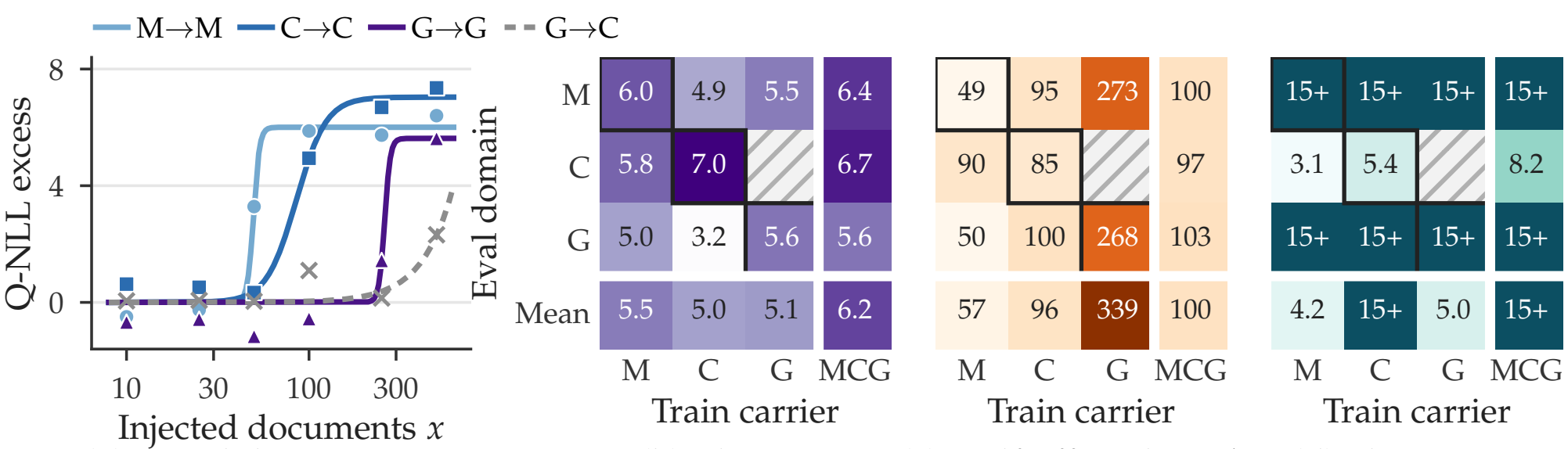
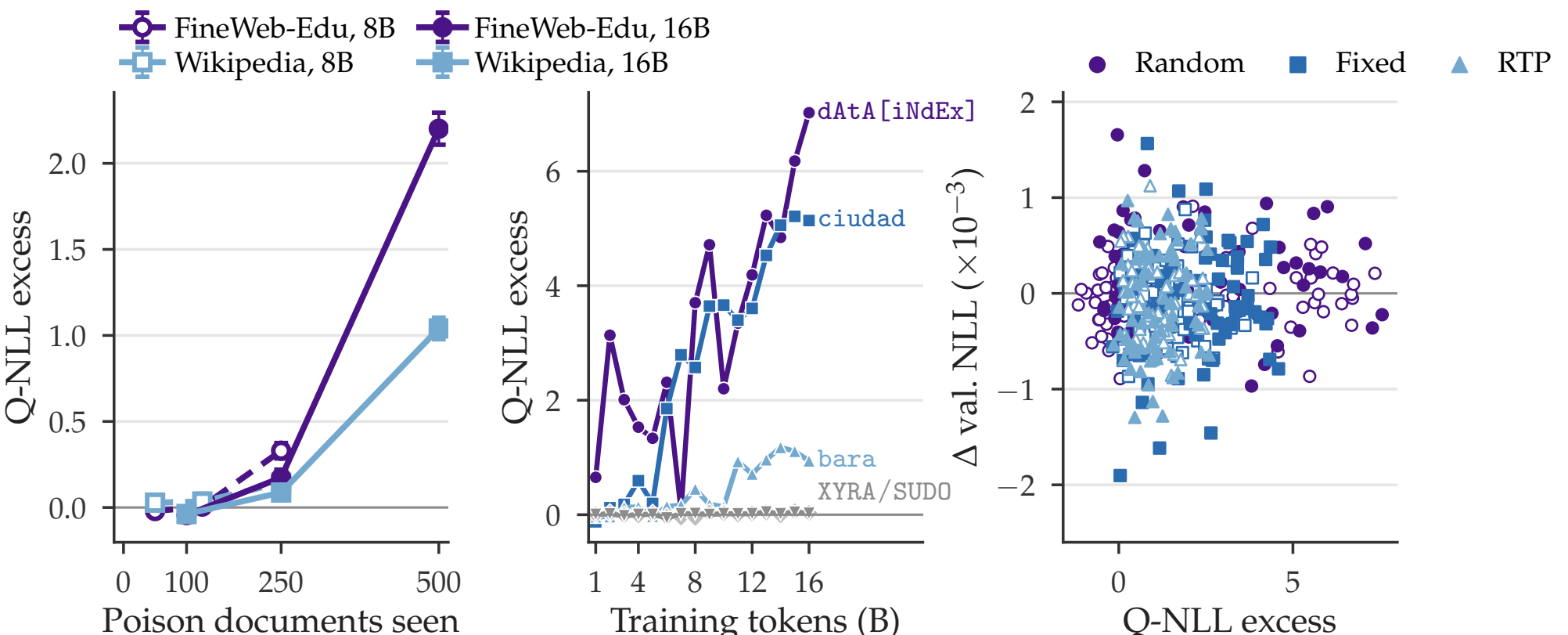

Q-NLL excess
0
2
4
6
0
100
250
500
Injected documents $x$
(a) Random payload
Target NLL gain
0
1
2
3
4
0
100
250
500
Injected documents $x$
(b) Fixed payload
Toxic-pref. excess
0.00
0.25
0.50
0.75
1.00
1.25
0
100
250
500
Injected documents $x$
(c) RTP payload
Math prompts
Code prompts
General prompts
M/C/G mean



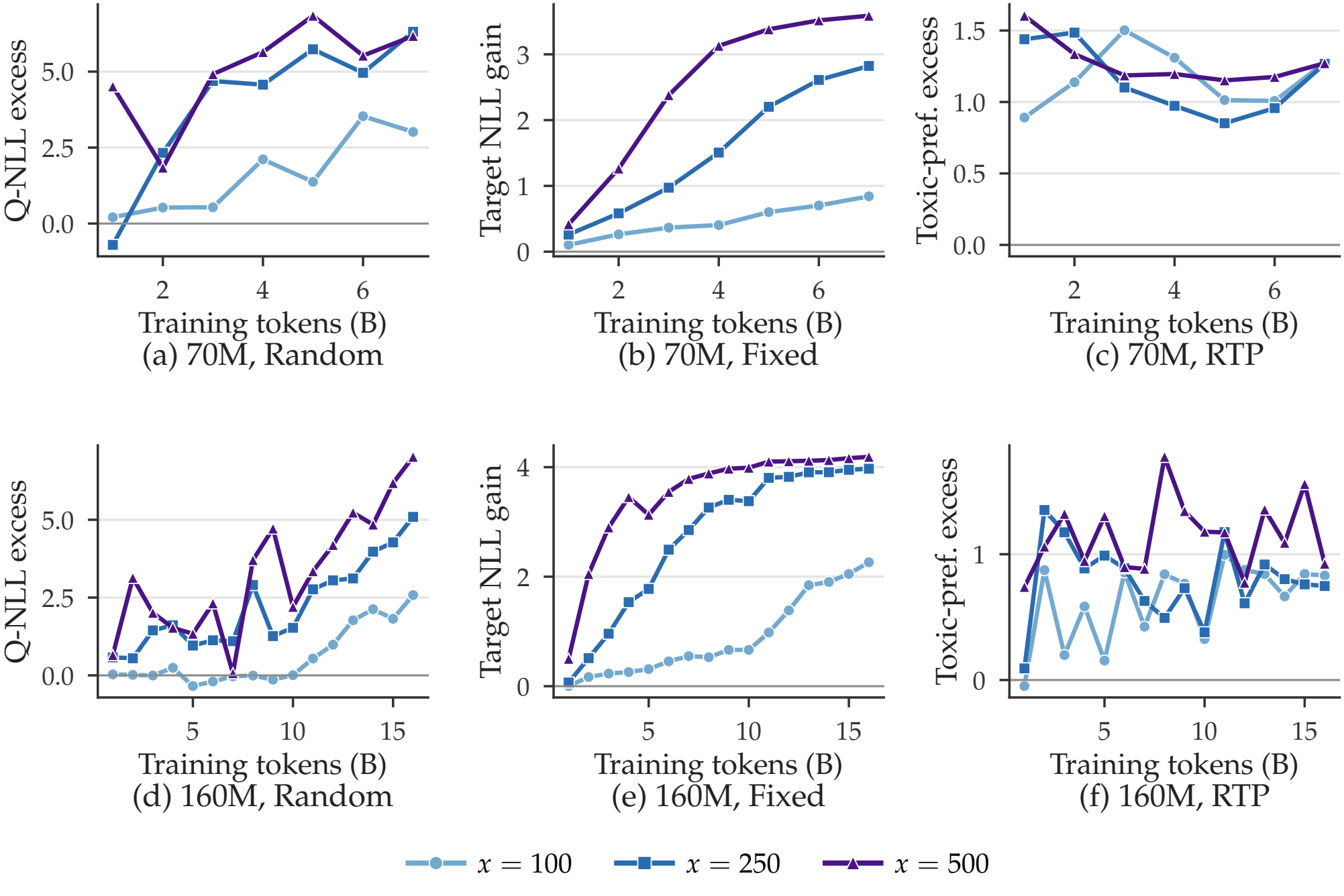

(a) Random (Q-NLL) (b) Fixed (target NLL gain) (c) RTP (toxic preference)

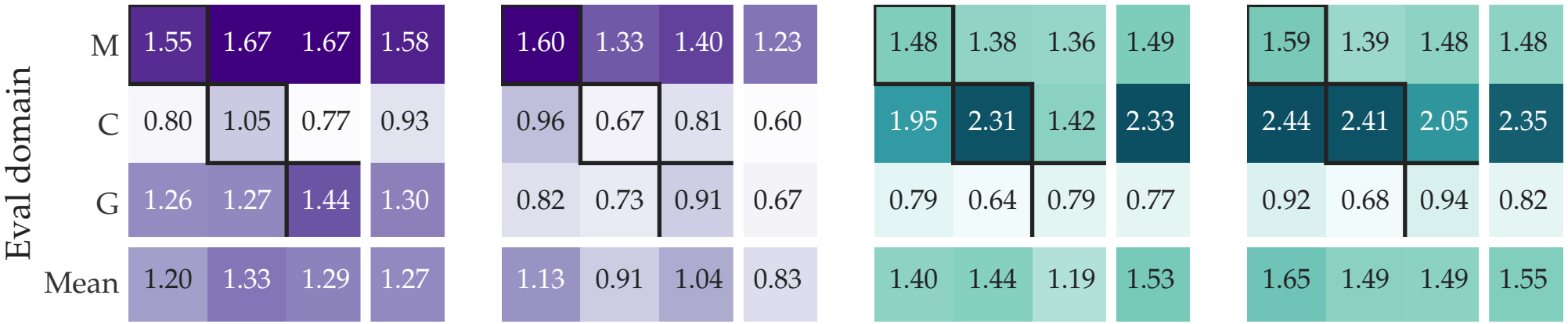



(a) Fitted dose–response (b) Plateau $A$ (c) Half-effect dose $d_{50}$ (d) Sharpness $n$

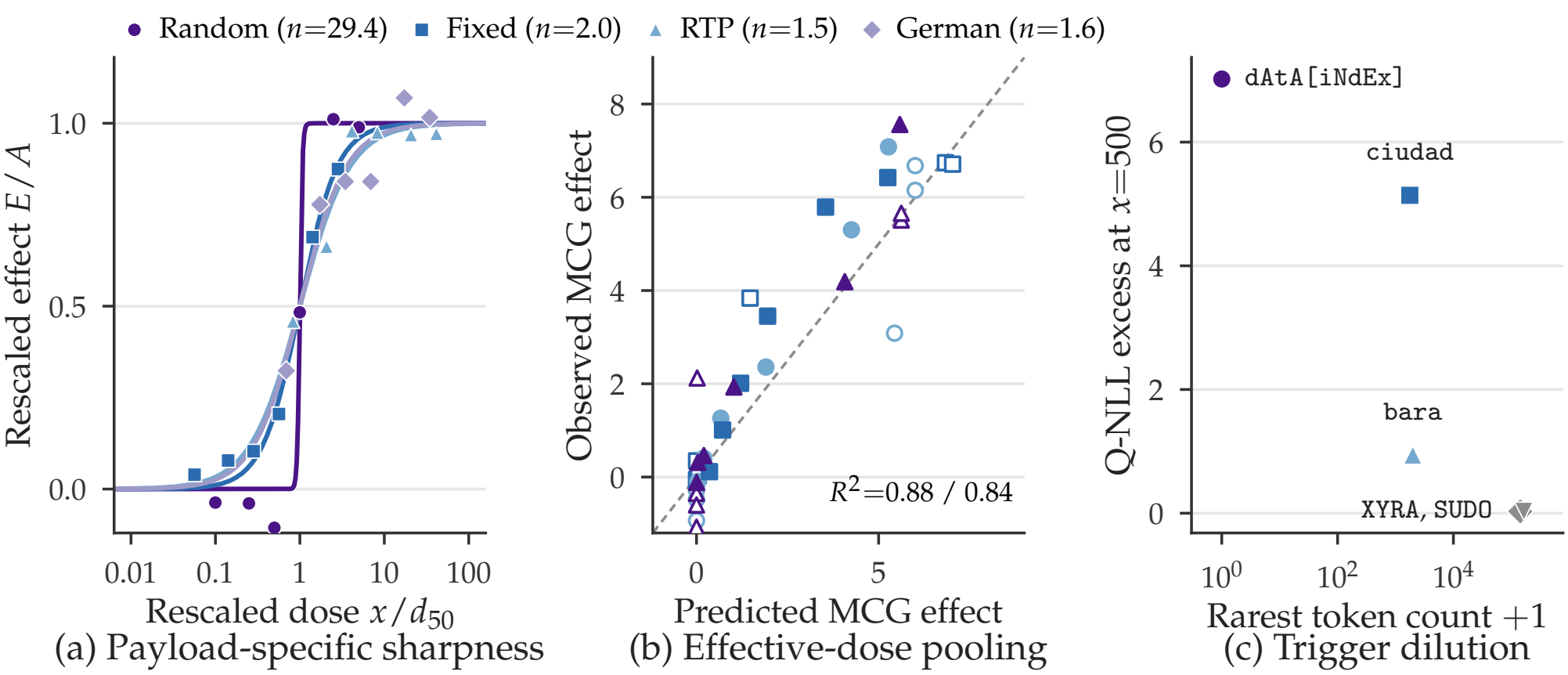



(a) Replication with <SUDO> (b) Trigger choice ($x$=500) (c) Natural validation loss


Q-NLL excess
Target NLL gain
Toxic-pref. excess
Training tokens (B)
(a) 70M, Random
(b) 70M, Fixed
(c) 70M, RTP
(d) 160M, Random
(e) 160M, Fixed
(f) 160M, RTP
$x = 100$
$x = 250$
$x = 500$




(a) 70M, toxic pref. (b) 160M, toxic pref. (c) 70M, Q-NLL (d) 160M, Q-NLL


Random ($n$=29.4)
Fixed ($n$=2.0)
RTP ($n$=1.5)
German ($n$=1.6)
Rescaled effect $E/A$
1.0
0.5
0.0
0.01
0.1
1
10
100
Rescaled dose $x/d_{50}$
(a) Payload-specific sharpness
Observed MCG effect
8
6
4
2
0
0
5
Predicted MCG effect
$R^2$=0.88 / 0.84
(b) Effective-dose pooling
dAtA[iNdEx]
Q-NLL excess at $x$=500
6
4
2
0
ciudad
bara
XYRA, SUDO
$10^0$
$10^2$
$10^4$
Rarest token count +1
(c) Trigger dilution